



Kanishk P 251801127-2025-AIDS ▾

K2**Started on** Thursday, 6 August 2026, 11:04 AM**State** Finished**Completed on** Thursday, 27 August 2026, 10:54 AM**Time taken** 20 days 23 hours**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int m;
5      scanf("%d",&m);
6
7      int a[m];
8
9      for(int i=0;i<m;i++)
10     {
11         scanf("%d",&a[i]);
12     }
13
14     int low=0;
15     int high=m-1;
16     int firstzero=m;
17     int mid;
18
19     while(low<=high)
20     {
21         mid=(low+high)/2;
22
23         if(a[mid]==0)
24         {
25             firstzero=mid;
26             high=mid-1;
27         }
28
29         else{
30             low=mid+1;
31         }
32     }
33
34     printf("%d",m-firstzero);
35 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Kanishk P 251801127-2025-AIDS](#) ▾**K2****Started on** Thursday, 6 August 2026, 11:21 AM**State** Finished**Completed on** Thursday, 27 August 2026, 11:07 AM**Time taken** 20 days 23 hours**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i;
5
6
7      scanf("%d", &n);
8
9      int nums[n];
10
11
12  for (i = 0; i < n; i++) {
13      scanf("%d", &nums[i]);
14  }
15
16  int left = 0, right = n - 1;
17
18  while (left < right) {
19      int mid = (left + right) / 2;
20      int candidate = nums[mid];
21
22      int leftCount = 0;
23      int rightCount = 0;
24
25      // Count candidate in left half
26  for (i = left; i <= mid; i++) {
27      if (nums[i] == candidate)
28          leftCount++;
29  }
30
31      // Count candidate in right half

```

```
31 // Count candidate in right half
32 for (i = mid + 1; i <= right; i++) {
33     if (nums[i] == candidate)
34         rightCount++;
35 }
36
37 // Check which half contains the majority candidate
38 if (leftCount > (mid - left + 1) / 2) {
39     right = mid;
40 }
41 else if (rightCount > (right - mid) / 2) {
42     left = mid + 1;
43 }
44 else {
45     left = mid + 1;
46 }
47 }
48
49 printf("%d\n", nums[left]);
50
51 return 0;
52 }
```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Kanishk P 251801127-2025-AIDS](#) ▾**K2****Started on** Thursday, 27 August 2026, 11:07 AM**State** Finished**Completed on** Thursday, 27 August 2026, 11:34 AM**Time taken** 26 mins 34 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d",&n);
7
8      int a[n];
9
10     for(int i=0;i<n;i++)
11     {
12         scanf("%d",&a[i]);
13     }
14
15     int x;
16     scanf("%d",&x);
17
18     int low=0;
19     int high=n-1;
20     int mid;
21     int floor=-1;
22
23     while(low<=high)
24     {
25         mid=low+(high-low)/2;
26
27         if(a[mid]==x)
28         {
29             floor=a[mid];
30             break;
31         }
32
33         else if(a[mid]>x)
34         {
35             high=mid-1;
36         }
37
38         else
39         {
40             floor=a[mid];
41             low=mid+1;
42         }
43     }
44
45     printf("%d",floor);
46 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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K2**Started on** Thursday, 27 August 2026, 11:34 AM**State** Finished**Completed on** Thursday, 27 August 2026, 11:39 AM**Time taken** 4 mins 45 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, x;
6      int arr[100];
7
8      scanf("%d", &n);
9
10     for (int i = 0; i < n; i++)
11     {
12         scanf("%d", &arr[i]);
13     }
14
15     scanf("%d", &x);
16
17     int found = 0;
18
19     for (int i = 0; i < n - 1; i++)
20     {
21         int target = x - arr[i];
22
23         /* Divide and conquer: Binary Search */
24         int low = i + 1;
25         int high = n - 1;
26
27         while (low <= high)
28         {
29             int mid = low + (high - low) / 2;
30
31             if (arr[mid] == target)
32             {
33                 printf("%d\n", arr[i]);
34                 printf("%d\n", arr[mid]);
35                 found = 1;
36                 break;
37             }
38             else if (arr[mid] < target)
39             {
40                 low = mid + 1;
41             }
42             else
43             {
44                 high = mid - 1;
45             }
46         }
47
48         if (found)
49             break;
50     }

```

51

52

if (!found)

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Kanishk P 251801127-2025-AIDS](#) ▾**K2****Started on** Thursday, 27 August 2026, 11:39 AM**State** Finished**Completed on** Thursday, 17 September 2026, 10:08 AM**Time taken** 20 days 22 hours**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      // Read the number of elements
6      if (scanf("%d", &n) != 1) return 0;
7
8      int arr[n];
9      // Read the elements
10     for (int i = 0; i < n; i++) {
11         scanf("%d", &arr[i]);
12     }
13
14     // Explicit stack to store low and high indices (avoiding recursion/functions)
15     int stack[1000];
16     int top = -1;
17
18     // Push initial low and high indices
19     stack[++top] = 0;
20     stack[++top] = n - 1;
21
22     // Process while stack is not empty
23     while (top >= 0) {
24         int high = stack[top--];
25         int low = stack[top--];
26
27         // Partitioning logic
28         int pivot = arr[high];
29         int i = low - 1;
30
31         for (int j = low; j < high; j++) {
32             if (arr[j] <= pivot) {
33                 i++;
34                 // Swap arr[i] and arr[j]
35                 int temp = arr[i];
36                 arr[i] = arr[j];
37                 arr[j] = temp;
38             }
39         }
40
41         // Place pivot in correct position
42         int temp = arr[i + 1];
43         arr[i + 1] = arr[high];
44         arr[high] = temp;
45
46         int p = i + 1;

```

```
47 |
48 | // Push left side sub-array bounds to stack if applicable
49 | if (p - 1 > low) {
50 |     stack[++top] = low;
51 |     stack[++top] = p - 1;
52 | }
```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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